Ch 2.2.3: Intro to classification

Lecture 9 - CMSE 381

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Mon, Sep 18, 2023

Announcements

Lec #	Date			Reading	Homeworks	***Quizzes*** *(Note: These are not announced until after they happen)*
3	Fri	Sep 1	Assessing Model Accuracy	2.2.1, 2.2.2	HW #1 Due	Quiz #1
	Mon	Sep 4	No class - Labor day			
4	Wed	Sep 6	Linear Regression	3.1		
5	Fri	Sep 8	More Linear Regression	3.1/3.2		Quiz #2
6	Mon	Sep 11	Even more linear regression	3.2.2	Hw #2 Due	
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8	Fri	Sep 15	Linear regression coding module			
9	Mon	Sep 18	Intro to classification, Bayes classifier, KNN classifier	2.2.3		
10	Wed	Sep 20	Logistic Regression	4.1, 4.2, 4.3.1-3		
11	Fri	Sep 22	Multiple Logistic Regression / Multinomial Logistic Regression /Project day	4.3.4-5	Hw #3 Due	
	Mon	Sep 25	***Review***			
	Wed	Sep 27	**Midterm #1**			
	Fri	Sep 29	No class - Dr Munch out of town			

Last Time:

- Finished Linear Regression

Announcements:

- Homework #3 Due Friday Sep 22
- Next Monday - Review day
  - Nothing prepped
  - Bring your questions
- Weds 9/27 - Exam #1
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in

Covered in this lecture

- Ch 2.2.3
- Error rate (classification)
- Bayes Classifier
- K-NN classification

Section 1

Classification Overview

What is classification

Classification: When the response variable is qualitative

- Given feature vector X and qualitative response Y in the set S, the goal is to find a function (classifier) $C(X)$ taking X as input and predicting its value for Y.
- We are more interested in estimating the probabilities that X belongs to each category

Y

Comes from an unordered set

Eye color $\in \{$	brown,	blue,	green,	orange $\}$
	1	0	0	0
prob	$1/2$	$\frac{1}{4}$	$1/8$	$1/8$

Some examples

- Predict whether a COVID19 vaccine will work on a patient given patient's age
- An online banking service wants to determine whether a transaction being performed is fraudulent on the basis of the user's IP address, past transactions, etc.

Section 2

Ch 2.2.3: Classification

Error rate

$x_i = (x_1^i, ___, x_p^i)$

- Training data: $\{(x_1, y_1), \cdots, (x_n, y_n)\}$ with y_i qualitative
- Estimate $\hat{y} = \hat{f}(x)$ Returns a label
- Indicator variable

$$I(\underbrace{y_i \neq \hat{y}_i}_{\text{true}}) = \begin{cases} 1 & \text{if not the same} \\ & \text{misclassified} \\ 0 & \text{else} \\ & \text{correctly classified} \end{cases}$$

Training error rate:

$$\frac{1}{n}\overbrace{\sum_{i=1}^{n} \mathrm{I}(y_i \neq \hat{y}_i)}^{\#\text{ of misclassifications}}$$

% of incorrect answers

Test error rate:

$$\underbrace{\mathrm{Ave}(\mathrm{I}(y_0 \neq \hat{y}_0))}$$

prob of being wrong on unseen data point

Best ever classifier

We can't have nice things

If I know the distribution used to simulate data this is the classifer where I'm wrong least often.

Bayes Classifier:
Give every observation the highest probability class given its predictor variables

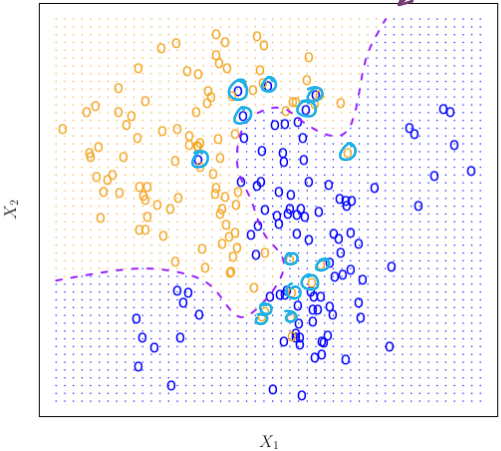
$$\Pr(Y = j \mid X = x_0)$$

An example

- Survey students for amount of programming experience, and current GPA
- Try to predict if they will pass CMSE 381.
- If we have a survey of all students that could ever exist, we can determine the probability of failure given combo of those features.

No single classifier will get it right 100% of the time because students with the same background might pass or fail.

Bayes decision boundary



Bayes decision boundary

$Y \in \{\text{blue}, \text{yellow}\}$

↳ Where we switch predictions

↳ Note this is more complex with more levels of Y

Bayes error rate

$Y \in \{1, \ldots, g\}$ Not ordered set! But there's g of them

- Error at $X = x_0$

$$1 - \max_j \underbrace{\Pr(Y = j \mid X = x_0)}_{\text{prob of seeing output } j}$$

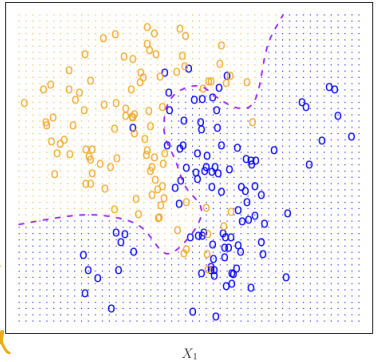
When I'm wrong b/c what happened was not the max prob label

- Overall Bayes error:

$$1 - E\left(\underbrace{\max_j \Pr(Y = j \mid X = x_0)}_{\text{When I'm right}}\right)$$

expected value

prob that I'm wrong



Analagous to irreducible error.

The game

I don't get to have Bayes classifier.

Goal: do as good a job as possible estimating the Bayes decision boundary

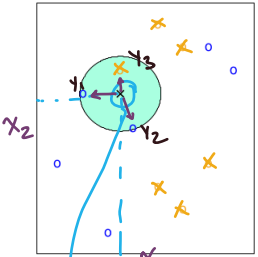
Classify given observation with highest estimated probability

Section 3

K-Nearest Neighbors Classifier

K-Nearest Neighbors

training data
blue / yellow



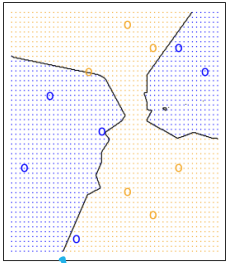
$K = 3$

New data point

- Fix K positive integer
- $N(x)$ = the set of K closest neighbors to x
- Estimate conditional proability

$$\Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in N(x_0)} \mathrm{I}(y_i = j)$$

- Pick j with highest value



Black line: KNN decision boundary

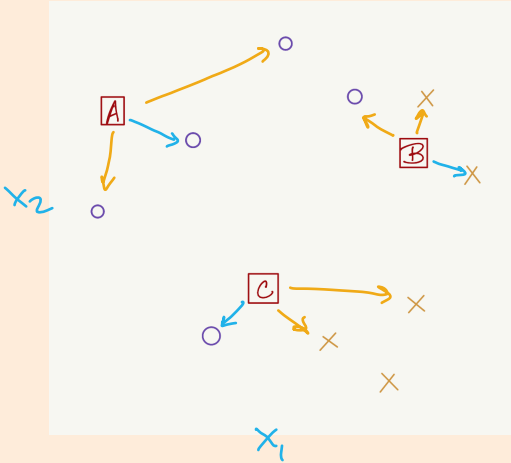
I will report blue

$$\Pr(Y = \text{blue} \mid X = x_0) = \frac{1}{3}\left(I(y_1 = \text{blue}) + I(y_2 = \text{blue}) + I(y_3 = \text{blue})\right) = \frac{1}{3}(1 + 1 + 0) = \frac{2}{3}$$

$$\Pr(Y = \text{yellow} \mid X = x_0) = \frac{1}{3}(0 + 0 + 1) = \frac{1}{3}$$

Example

Here label is shown by O vs X. What are the knn predictions for points A, B and C for $k = 1$ or $k = 3$?

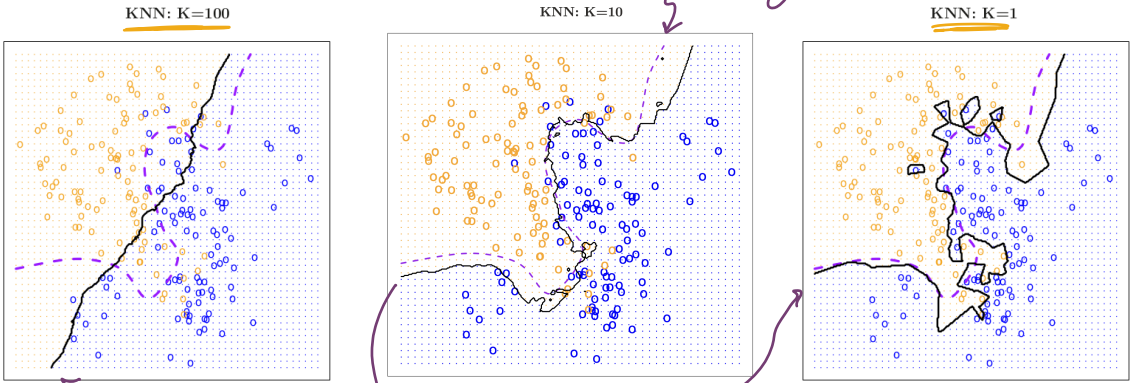


Point	$k = 1$ Prediction	$k = 3$ Prediction
A	O	O
B	X	X
C	O	X

Tradeoff

K controls flexibility

purple dash is Bayes decision bdy

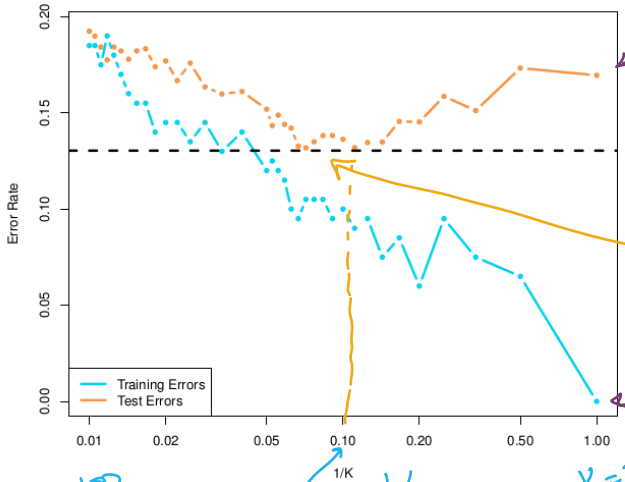


Not enough flexibility

Black line: KNN decision boundary

Boundary is overly flexible

More on tradeoff



Test error

U-shape

What k do I want?

training error

k=100

k=10

1/K

k=1

Jupyter notebook

Next time

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